

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location Digital Realty Northern Virginia IAD44, VA -- station KIAD, America/New_York

Committed pair OSM 1317733642 -> 1543296494
Digital Realty Northern Virginia IAD44 / Digital Realty Northern Virginia IAD44

Facade gap 63.8 m between the two halls

Weather record 43,763 real hourly records from KIAD

Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.4548 C at 295 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2025-03-11 (crossing)

Plant limit 18.0 C

Notice required 3 h

Forecast skill 0.50 relative to persistence

Level anchor sensor

Condenser bank longest facade

Switch budget 1 mode changes per day

Minimum dwell 3 h

Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 2 of 24 hours with 1 mode change. The reactive incumbent took 13 hours -- 11 more -- but declared 2 unsafe hours against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 12 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 2 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)

Incumbent 13 of 24 hours free cooling, 2 mode change(s), 2 unsafe hour(s)
the incumbent broke its own switch budget 1 time(s) to stay safe
12 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	4.599	18.0	4.182	1.272
01	mechanical	yes	switch budget	4.599	18.0	4.182	1.213
02	mechanical	yes	switch budget	4.319	18.0	3.072	1.382
03	mechanical	yes	switch budget	3.165	18.0	1.554	1.254
04	mechanical	yes	switch budget	3.214	18.0	0.852	1.211
05	mechanical	yes	switch budget	2.629	18.0	0.292	1.387
06	mechanical	yes	switch budget	1.269	18.0	-0.818	1.202
07	mechanical	yes	switch budget	2.934	18.0	0.292	1.855

08	mechanical	yes	switch budget	6.406	18.0	5.727	2.084
09	mechanical	yes	switch budget	10.274	18.0	11.522	2.577
10	mechanical	yes	switch budget	12.118	18.0	14.307	2.413
11	mechanical	yes	switch budget	16.088	18.0	18.052	2.001
12	mechanical	no	dry-bulb	18.815	18.0	19.852	1.540
13	mechanical	no	dry-bulb	20.173	18.0	21.550	1.391
14	mechanical	no	dry-bulb	22.535	18.0	22.119	1.512
15	mechanical	no	dry-bulb	22.840	18.0	23.187	1.282
16	mechanical	no	dry-bulb	23.492	18.0	23.187	1.320
17	mechanical	no	dry-bulb	23.321	18.0	22.627	1.614
18	mechanical	no	dry-bulb	22.690	18.0	21.517	1.505
19	mechanical	no	dry-bulb	21.391	18.0	19.302	1.620
20	mechanical	no	dry-bulb	20.517	18.0	18.052	1.745
21	mechanical	no	dry-bulb	18.985	18.0	15.946	1.831
22	FREE	yes	-	16.340	18.0	12.650	1.557
23	FREE	yes	-	16.303	18.0	13.610	1.350

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.599 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.599 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.319 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.165 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.214 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.629 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 1.269 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.934 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 6.406 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.274 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.118 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.088 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.815 C against a 18.0 C limit, so it fails by 0.815 C. It would take a limit 0.815 C higher, or a bound 0.815 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.173 C against a 18.0 C limit, so it fails by 2.173 C. It would take a limit 2.173 C higher, or a bound 2.173 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.535 C against a 18.0 C limit, so it fails by 4.535 C. It would take a limit 4.535 C higher, or a bound 4.535 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.840 C against a 18.0 C limit, so it fails by 4.840 C. It would take a limit 4.840 C higher, or a bound 4.840 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.492 C against a 18.0 C limit, so it fails by 5.492 C. It would take a limit 5.492 C higher, or a bound 5.492 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.321 C against a 18.0 C limit, so it fails by 5.321 C. It would take a limit 5.321 C higher, or a bound 5.321 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.690 C against a 18.0 C limit, so it fails by 4.690 C. It would take a limit 4.690 C higher, or a bound 4.690 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.391 C against a 18.0 C limit, so it fails by 3.391 C. It would take a limit 3.391 C higher, or a bound 3.391 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.517 C against a 18.0 C limit, so it fails by 2.517 C. It would take a limit 2.517 C higher, or a bound 2.517 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.985 C against a 18.0 C limit, so it fails by 0.985 C. It would take a limit 0.985 C higher, or a bound 0.985 C tighter, to change this hour.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.340 C, which is 1.660 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.557 C, and the margin is measured, not chosen: 1.189 C of group-conditional forecast error for this hour of day, plus 0.3682 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.303 C, which is 1.697 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.350 C, and the margin is measured, not chosen: 0.953 C of group-conditional forecast error for this hour of day, plus 0.3965 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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